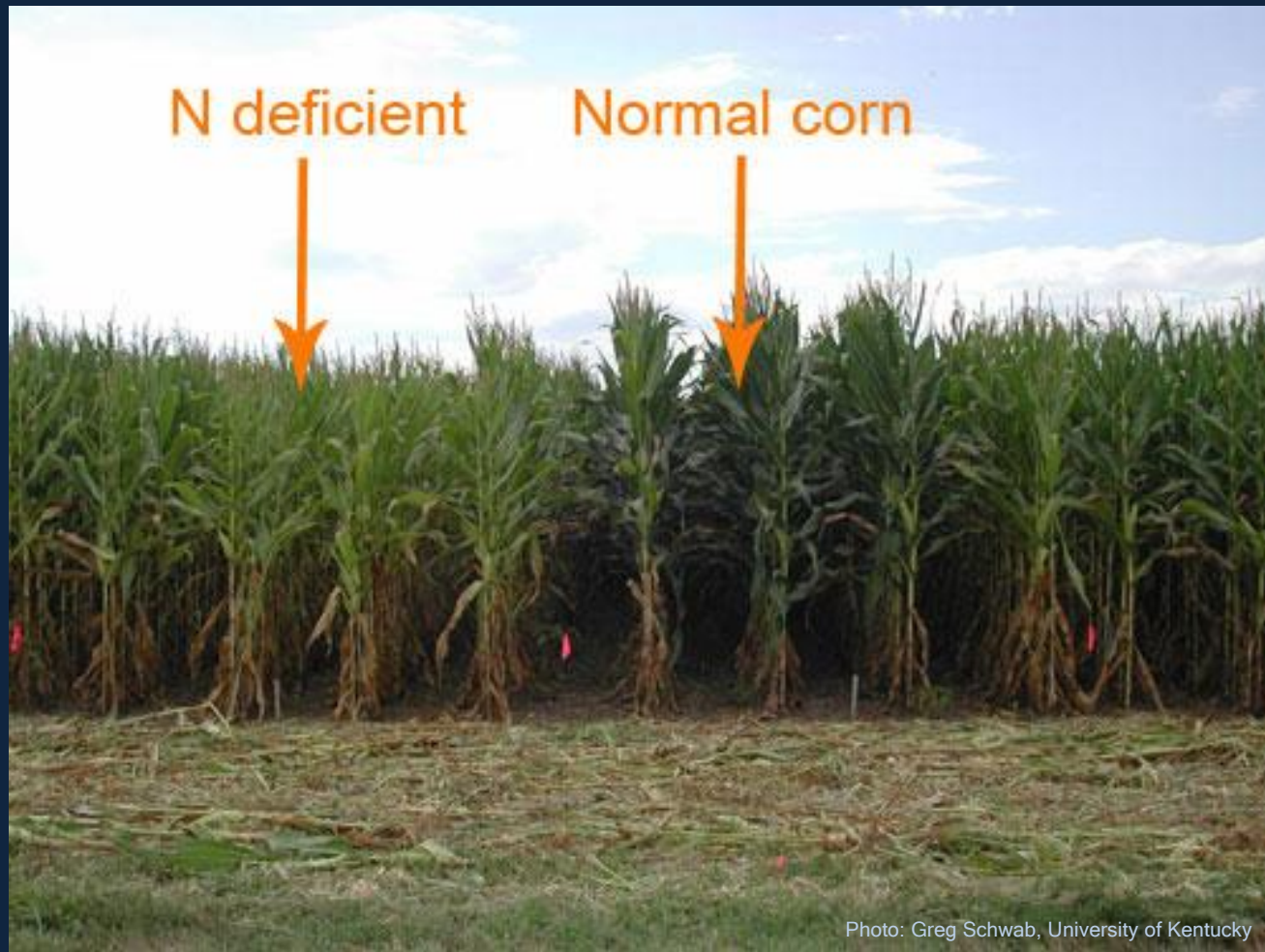




Nitrogen deficiency in corn. On the left is a normal green leaf blade. Others show progressive nitrogen deficiency symptoms. Yellowing of the leaf tissue occurs first in the older leaves and follows the midrib from the leaf tip, showing a characteristic V-shaped wedge. Later the tip begins to dry and the whole leaf may be involved, an effect known as 'firing'.



Under field conditions N deficiency in corn produces stunted, light yellowish-green plants, as compared to normal corn. In this photo it's easy to see the difference between the N deficient corn and the normal corn. Plants may also suffer from mild N deficiency where symptoms, to the untrained eye, are not as striking. Growers must estimate crop N demand (the 'agronomic N rate') and then supply that amount of N to the plant.





Nitrogen is mobile in the plant, thus symptoms of N deficiency show up first on the bottom leaves.



Photos: Mosaic Corporation <http://back-to-basics.net/nds/index.htm>

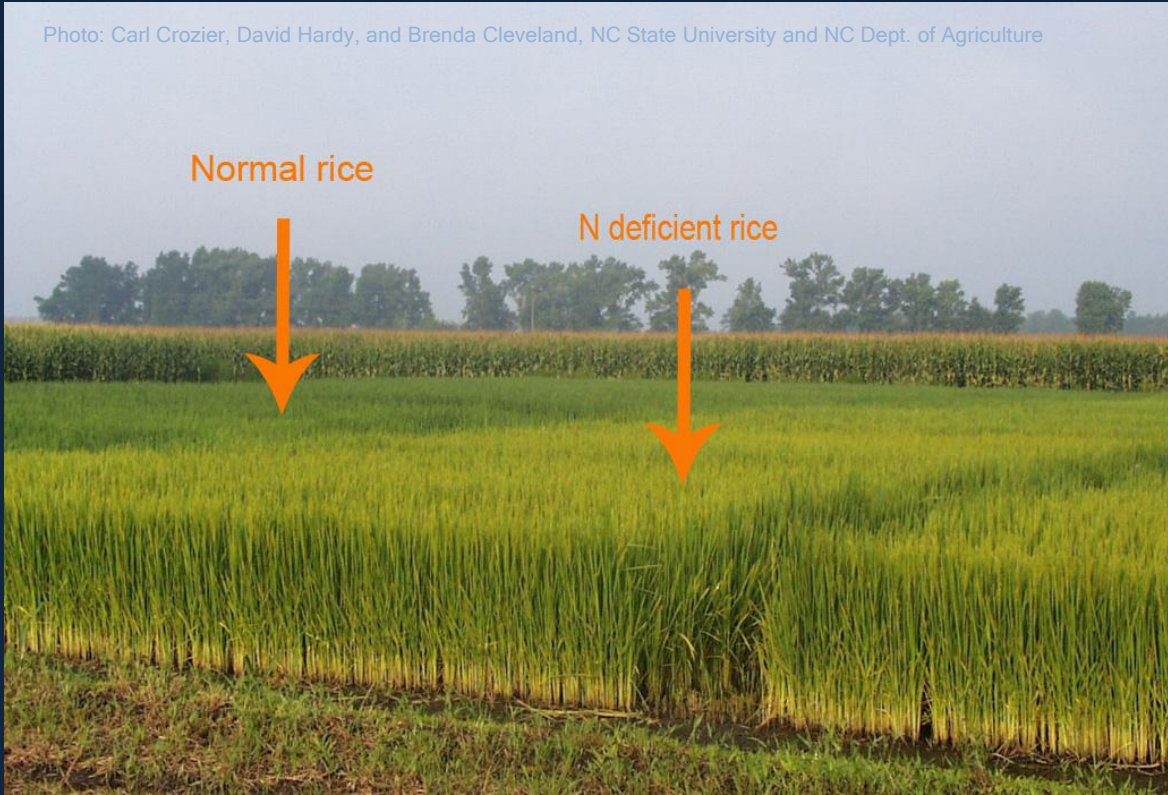


Cotton plants deficient in nitrogen produce small, pale yellow leaves, reduced stalk height, and fewer vegetative branches. Bolls are shed in the first 10-12 days after flowering. Nitrogen deficiency occurring later in the season results in yellowing, premature reddening and early shedding of leaves.

Photo: Carl Crozier, David Hardy, and Brenda Cleveland, NC State University and NC Dept. of Agriculture

Normal rice

N deficient rice



Without a chemical test, nutrient deficiencies in plants cannot be diagnosed with certainty. Nitrogen deficiency in rice may show symptoms of yellowish-green tinted plants but not much difference in height. At right, a healthy green leaf of wheat (*a*) showing balanced nutrition. Yellowish-green leaf with tip end turning yellow (*b*) indicates N deficiency. Compare these to (*c*) and (*d*), phosphorus and potassium starvation, respectively. These symptoms are diagnosed with difficulty under field conditions. Further complicating things, disease organisms and environmental stress may induce look-alike symptoms.





Image: American Society of Agronomy

Shoots and leaves of tomato plants grown under conditions of N starvation. The older mature leaves gradually change from their normal dark green color to a paler green color, shown in the specimen on the right. The younger leaves retain their green color but tend to be smaller in size. The specimen on the left shows acute N deficiency wherein older leaves fade to golden-yellow, and reddish or purple midribs and petioles. Lack of green color in plants is called *chlorosis*, a symptom that is diagnostic of N deficiency as well as other non-fertility related disorders. Recovery of deficient plants to applied nitrogen is immediate (days).

Image: American Society of Agronomy

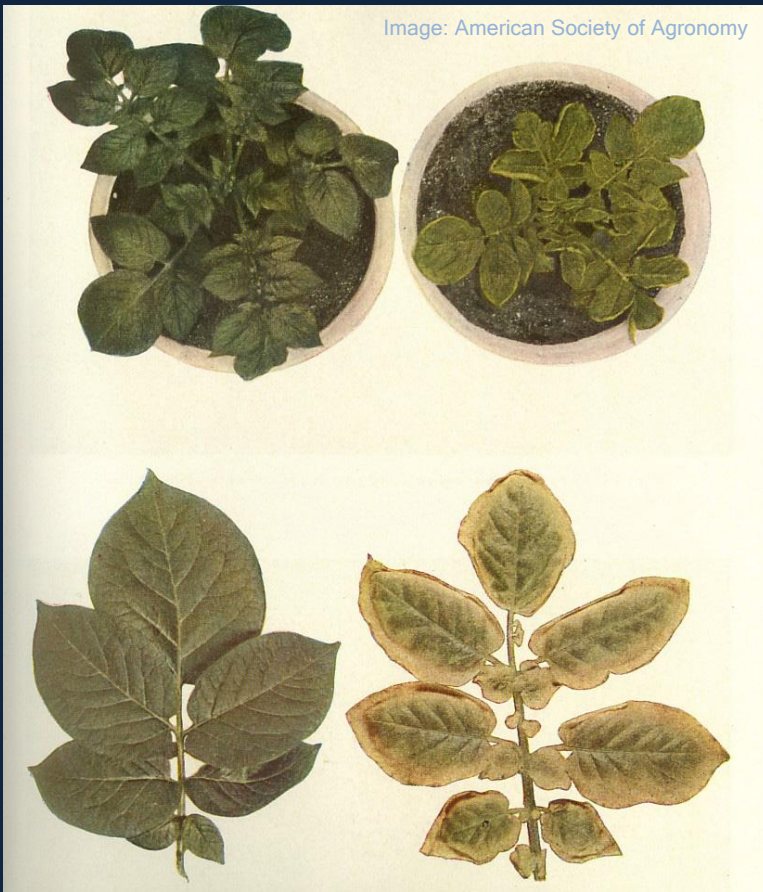
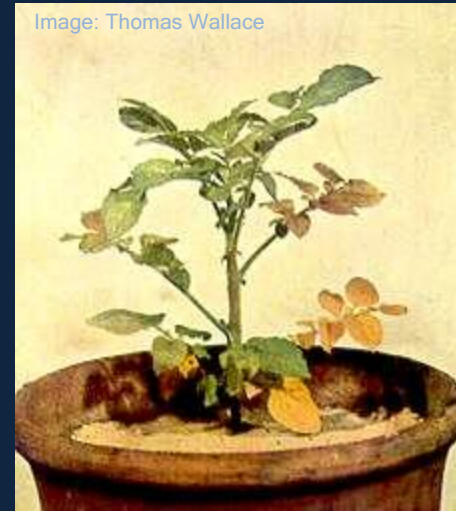


Image: Thomas Wallace



Above: The potato plant in the plot showing yellowing of the older leaves characteristic of N deficiency.

Above: The potato plant in the pot at the left is normal. The one on the right shows the beginning stages of N deficiency, of which uprolling and cupping of the leaflets and general reduction in size and vigor are characteristic symptoms. Below: The leaf on the left is normal. The leaf at the right is from a potato plant growing in soil highly deficiency in N. The distinct curling and 'scorching' of the leaf margins are characteristic symptoms. Note that scorching of leaf margins is also a characteristic symptom of potassium shortage. A chemical test of the leaf petiole will diagnose the problem.